

Article

Investigating Land Cover Changes and Their Impact on Land Surface Temperature in Khyber Pakhtunkhwa, Pakistan

Hammad Ul Hussan ^{1,2}, Hua Li ^{1,2,*} , Qinhua Liu ^{1,2} , Barjeece Bashir ^{1,3} , Tian Hu ⁴ and Shouyi Zhong ⁵

¹ Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China; chhmaad1@mails.ucas.ac.cn (H.U.H.); liuqh@aircas.ac.cn (Q.L.); barjeece@radi.ac.cn (B.B.)

² University of Chinese Academy of Sciences, Beijing 100049, China

³ International Research Center of Big Data for Sustainable Development Goals, Beijing 100094, China

⁴ Luxembourg Institute of Science and Technology, 4362 Belvaux, Luxembourg; tian.hu@list.lu

⁵ Faculty of Geographical Science, Beijing Normal University, Beijing 100875, China; zhongsy@mail.bnu.edu.cn

* Correspondence: lihua@aircas.ac.cn

Abstract: Restoration of degraded land is a significant concern in the 21st century in order to combat the impacts of climate change. For this reason, the provisional government of Khyber Pakhtunkhwa (KPK), Pakistan, initialized a Billion Tree Tsunami Project (BTTP) in 2013 and finished it in 2017. Although a few researchers have investigated the land use transitions under BTTP in the short term by merging all the vegetation types into one, analysis of the long-term benefits of the project and future persistence were missing. Furthermore, the previous studies have not discussed whether the prime objective of the BTTP was achieved. Considering the existing gaps, this research mainly involves analyzing (i) fluctuations in the green fraction by employing a land change modeler (LCM), along with the spatial location of gain-loss and exchange analysis using a high-resolution dataset (GLC30); (ii) forest cover changes under the influence of the BTTP; (iii) impacts of green fraction changes towards land surface temperature (LST) by utilizing the less-explored technique of curve fit linear regression modeling (CFLR); and finally, (iv) assessing the persistence of the NDVI and LST trends by employing the Hurst exponent. Research findings indicate that as an output of BTTP, despite the government's claim of increasing the forest cover by 2%, a significant gain of grassland (3904.87 km²) was observed at the cost of bare land. In comparison, the overall increase in forest cover was only 0.39%, which does not satisfy the main objective of this project. On the other hand, the CFLRM-based actual contributions of land cover change (LCC) transition to LST indicate a significant decline in LST in the areas with gains in green fraction for both grassland and forest. At the same time, an increase was observed with reverse transitions. Although the results appear positive for climatic impacts in the short term, the HURST model-based persistence analysis revealed that the spatial locations of increasing vegetation and decreasing LST trends fall under the weakly persistent category, therefore these trends may not continue in the near future. Despite some positive impact on LST attributed to the green fraction increase, this project cannot be regarded as a complete success due to its failure to achieve its prime objective.

Keywords: land surface temperature; land cover change; climate change; Billion Tree Tsunami Project; Hurst exponent



Citation: Ul Hussan, H.; Li, H.; Liu, Q.; Bashir, B.; Hu, T.; Zhong, S. Investigating Land Cover Changes and Their Impact on Land Surface Temperature in Khyber Pakhtunkhwa, Pakistan. *Sustainability* **2024**, *16*, 2775. <https://doi.org/10.3390/su16072775>

Academic Editors: Daniela Ducci, Fabrizio Carteni, Luigi Rosati, Federica Dell'Acqua and Stefania Stevenazzi

Received: 26 February 2024

Revised: 18 March 2024

Accepted: 22 March 2024

Published: 27 March 2024



Copyright: © 2024 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

1. Introduction

The pace of land cover changes (LCCs) has surged four times in the last few decades [1] as the global urban population continuously increases (51% in 2010) and is estimated to reach 80% of global population by the year 2050 [2]. The upsurge in urban population puts pressure on resources. It causes land cover changes within a specific area, as an increase in population requires several facilities, e.g., housing, industry, roads, and other infrastructure, causing severe LCC transitions. In the last few decades, the LCC transitions have been

happening at a macro scale, in which forest cover has been transformed into cultivated land, and cultivated land has changed into artificial surfaces to cater to human needs [3]. On the other hand, an unprecedented proliferation of artificial surfaces, intensification in agriculture, and significant alteration of the green cover fraction can play a vital role in altering ecosystems, e.g., biodiversity loss, soil erosion, water cycle disruption [4], loss of ecosystem services, disruption of ecological relationships [5], and alternating land surface temperature (LST) [6,7]. Recently, several studies [1,8,9] have investigated the effects of land cover changes on the global environment, especially on LST [10], as variations in land cover categories can significantly influence LST [11–13]. Each land cover type makes a different contribution to the LST as every land cover type possesses a unique surface value of reflectance, coarseness, and distinctive qualities in terms of energy radiation and absorption [14–17]. To investigate these impacts and relations, remote sensing data has been widely used with acceptable accuracy for LST estimations [18,19] and LCC investigations [20]. Although direct observations from local meteorological stations provide more accurate data with higher temporal resolution as compared to remote sensing, these data have limitations such as expensiveness, data unavailability for remote areas, and also the requirement for different statistical methods to first be applied before becoming useful for further studies [21,22]. On the other hand, remote sensing data is mostly cost-free and has continuously provided high spatial resolution imageries for diverse applications regarding environmental aspects since 1972 [18,20,23]. This continuity of the data and the online data analysis platform Google Earth Engine (GEE) provides the facility to analyze big data on cloud servers, eliminating the inconvenience of downloading the data onto local disks [24].

Recently, numerous studies have used remote sensing approaches to investigate environmental problems. Since the late 19th century, the global mean surface temperature has been increasing at an average rate of 0.08 °C per decade [25–27]. Additionally, according to the sixth assessment report (AR6) A.1.2 of the Inter-Governmental Panel on Climate Change (IPCC), the worldwide average temperature was 0.99 °C higher in the first two decades (2001–2020) of the 21st century as compared to the period from 1850 to 1900. To be precise, a 1.59 °C increase in temperature was observed on the land masses and a 0.88 °C increase was observed over the ocean surface [28]. Similarly, some other researchers, e.g., Xiao et al. [29], reported a 0.5 °C increase in LST in Guizhou, China, in three years due to a significant reduction in green cover. Hamdi et al. [30] have argued that the LST of Belgium has been rising at the rate of 0.15 °C for every decade since 1960 due to urbanization. Ovalle et al. [31] have reported an increase of approximately 11 °C in LST due to land cover changes (LCCs) in Luis Potosi, Mexico, in the last two decades. Using the Landsat series data, Jalan et al. [32] noted that the LST has increased from about 2.9 °C to 4.0 °C in Jaipur, India. Rehman et al. [21] have reported a 6 °C increase in LST due to global warming and loss of green fraction in Sindh Creeks.

To address the rising global warming problem, in 2015, 195 nations signed the Paris Agreement under the United Nations Framework Convention on Climate Change (UNFCCC) to reduce global warming to below 2 °C and, preferably, 1.5 °C [33]. Additionally, to tackle global warming issues, several nations initialized afforestation projects, e.g., the Great Green Wall (Africa) [34], China's Three-North Shelterbelt Project [35], the Bonn Challenge (Global) [36], the European Union's Green Deal, Australia's 20 Million Trees Program, 20 × 20 Initiative (Latin America and the Caribbean) [37], and many more have been proved successful at the cost of bare land. Following this, Pakistan ranks 5th on the list of most populated countries, with 241.49 million inhabitants [38], and 8th on the list of countries most susceptible to the effects of climate change [39]. Moreover, Pakistan is among the least-wooded countries, with forests covering less than 5% of its land area [40]. Additionally, according to the Forest Department of Pakistan, Pakistan has lost 25% of natural forest cover in the last few decades, with an annual rate of loss of more than 2% [41]. The country's natural forests are subjected to deforestation due to the transformation of cultivated land, overgrazing, and the acquisition of wood fuel [42]. To cope with these

scenarios, in 2013, the provincial government of Khyber Pakhtunkhwa (KPK), Pakistan, launched an afforestation project called the “Billion Tree Tsunami Project (BTTP)” to increase forest cover by 2% by planting one billion trees in the region, with the prime objective of reducing land degradation and restoring previously degraded land by the year 2020 [43]. This project was completed and achieved the required objectives in 2017 [44], which dramatically reshaped the region’s land cover scenario.

As the project was completed in 2017, it attracted researchers to investigate its impacts on climate. Mumtaz et al. (2023) have evaluated the effects of BTTP on carbon emission and temperature [39]. Khan et al. (2021) have assessed the success of plantation activities and the survival of vegetation in the Malakand forest division (a small proportion of the BTTP) [45]. Abbas et al. (2023) examined three different forest sites after the first phase of the BTTP and reported that the forest plantation and protected natural regeneration performance remained variable across these three regions, with moderate growth in the fraction of vegetation cover (FVC) [46]. Tariq et al. (2022) reported a 12.7% increase in vegetation cover in Peshawar under the BTTP [47]. Mumtaz et al. (2020) reported a decrease in LST with an increase in vegetation over Peshawar city [13]. However, these studies have discussed the overall green fraction changes and their impacts by merging all the classes into a single class (vegetation) instead of the separate evaluation of each vegetation class to assess the success of the BTTP. Furthermore, investigation of the long-term benefits of the project remains untouched. Hence, considering the limitations mentioned, we have analyzed in detail the land cover changes by further separating the vegetation into four classes: forest cover, cultivated land, shrubland, and grassland, to understand better which type of vegetation is increased after the BTTP. The current study has utilized a high-resolution dataset (GLC30) to study land cover changes in KPK province, Pakistan, from 2000 to 2020. Considering the literature on LCCs’ impacts on LST and the BTTP, the current investigation intended to (1) monitor the land cover changes under BTTP over the region in the last two decades by utilizing a land change modeler (LCM), (2) quantify the influence of each LCC transition on LST variations by applying the cutting-edge curve fit linear regression (CFLR) model, (3) discover vegetation and LST trends by employing a linear regression model, and (4) evaluate vegetation and LST persistence by employing the Hurst exponent. The findings of this research will provide details about the success of BTTP and lead to detailed guidelines, which are expected to be helpful for policymakers in further afforestation projects to restore Forest cover and degraded land.

2. Study Area and Data

2.1. Study Area

Khyber Pakhtunkhwa (Figure 1), commonly referred to as either KP or KPK, is the third largest province in Pakistan with a total population of 40.85 million, and the fourth most extensive by land area at 101,741 km² [38]. It is geographically located at 34.9526° N, 72.3311° E, with a varied landscape ranging from rugged mountain ranges, valleys, plains surrounded by hills, undulating submontane areas, and dense agricultural farms making the average elevation 2135 m (masl) with a minimum and maximum elevation of 170 m and 7856 m respectively (<https://en-gb.topographic-map.com/map-cpsdn/Khyber-Pakhtunkhwa/>) (accessed on 9 May 2023). These mountains exhibit a complex geological history, with thrust faulting, folding, and extensive glaciation shaping the landscape over millions of years. The region also features vast sedimentary basins, such as the Potwar Basin, which holds significant hydrocarbon reserves. Furthermore, KPK is marked by deep river valleys, high plateaus, and steep slopes, with the Indus River serving as a prominent geological feature, carving through the landscape and depositing fertile alluvial plains [48]. Regarding climate characteristics, KPK is categorized as a humid subtropical climate (Cfa) in the Köppen climate classification system. The average temperature ranges from 7.11 °C to 32.42 °C; however, the minimum and maximum temperatures can differ significantly depending on the specific season and area of the province due to the diversity of topography and elevation (<https://en.climate-data.org/asia/pakistan/khyber-pakhtunkhwa-2239/>)

(accessed on 14 November 2023). The provincial government of KPK embarked on the BTTP, an afforestation initiative, in 2013 [43], precipitating substantial LCCs over the past decade. Concurrently, the province's population surged from 17.7 million to 30.5 million from 1998 to 2017 (40.85 million 2023) [38], an increase of more than double. This demographic upsurge has exerted significant pressure, inducing notable changes in land cover across the region. These factors collectively motivated our selection of this region for in-depth study.

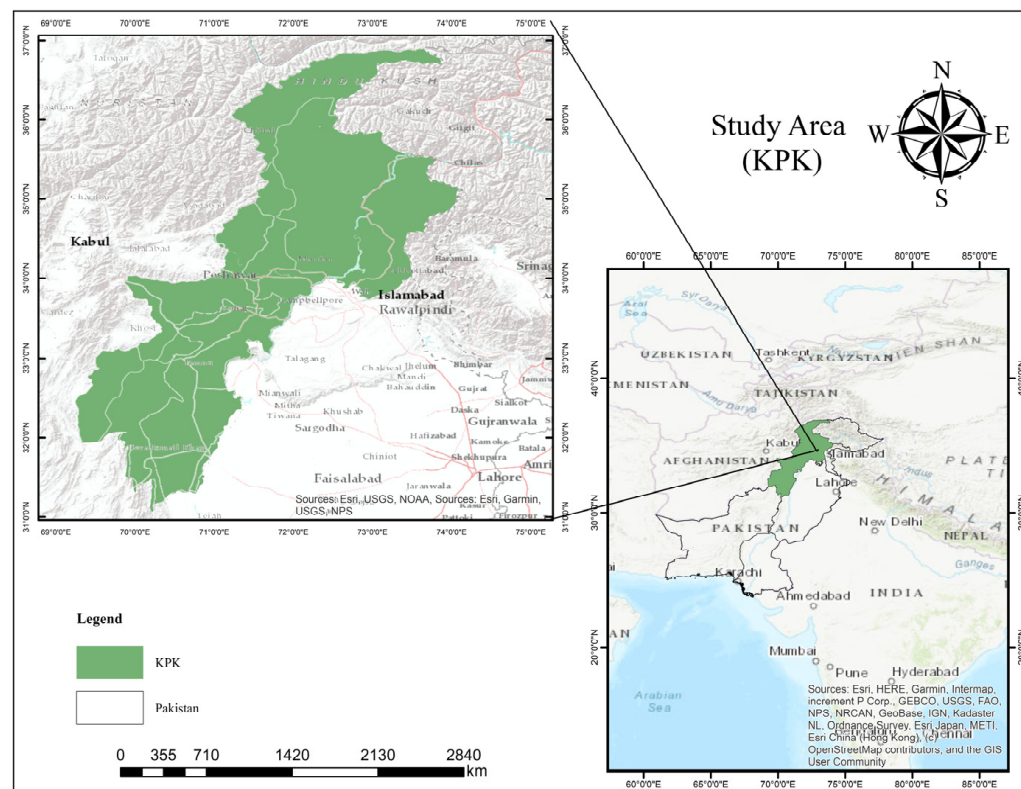


Figure 1. Study Area map (KPK).

2.2. Land Cover Data

To achieve the predetermined objectives, the current study employed high-resolution (30 m) datasets, including the “Global land cover-GLC30” dataset (<https://globeland30.org>) (accessed on 21 December 2023) which was generated using multi-spectral images from Landsat 5 Thematic Mapper (TM), Landsat 7 Enhanced Thematic Mapper Plus (ETM+), Landsat 8 Operational Land Imager (OLI), HuanJing (HJ)-1, and Gaofen (GF)-1. The product provides global land cover data for 2000, 2010, and 2020 with an overall accuracy ranging from 83.50% to 85.72% and a Kappa coefficient of 0.78–0.82. GLC30 includes 10 land cover classes in total, namely cultivated land, forest, grassland, shrubland, wetland, water bodies, tundra, artificial surface, bare land, and perennial snow and ice. To validate the land cover classification results of this product for the KPK region, 270 points (30 for each land cover type) were randomly plotted to assess the accuracy using Google Earth. The overall accuracy was calculated utilizing the Landsat imagery.

2.3. Landsat Data

The joint National Aeronautics and Space Administration (NASA)/U.S. Geological Survey (USGS) program, Landsat series, has been providing continuous high spatial resolution (30–120 m) data since 1972, which was made freely available to the public in 2008 [48]. In this study, we have utilized thermal observations from Landsat 5 Thematic Mapper (TM) and Landsat 8 Thermal Infrared Sensors (TIRs). Thermal observations from Landsat 5 TM (band 6) and Landsat 8 TIRs (band 10 and band 11) were originally at 60 m

and 100 m spatial resolution, respectively, which were resampled to 30 m. The mean seasonal imageries for the corresponding years were used to retrieve the LST, i.e., summer (June–September) and winter (December–February). Further, for the Hurst model, NDVI data for the growing season (May–November) was utilized with a <10% cloud cover filter in JavaScript in Google Earth Engine (GEE).

3. Methods

3.1. Land Cover Change and Transitions

To prepare the LCC transition matrices, the data of GLC30 was reclassified into nine classes, including artificial surfaces, bare land, forest, grassland, shrubland, cultivated land, wetland, water bodies, and permanent ice and snow, according to the first level classification standard of Land Use Status Classification (GB/T2010-2017 [39]), and the tundra class was merged with the shrubland category. To further validate the land cover change (LCC) classifications, we have applied ground truth data in ArcGIS 10.4 which was acquired from Google Earth to calculate the overall accuracy, and this revealed an overall accuracy of more than 85% for all LCC classes. Further, the LCC maps were generated by employing the land change modeler (LCM) in IDRISI 2020, a specialized tool that facilitates the analysis and modeling of temporal land cover change [49–51]. LCM was utilized to evaluate the rate and spatial location of gains and losses, transitions, and exchanges in LCC over KPK in the last two decades (2000–2020).

3.2. Land Surface Temperature Retrieval

This research utilized the thermal band (B6) of Landsat 5 TM and Band 10 of the Landsat 8 TIRS for the estimation of summer and winter mean LST. All the necessary radiometric correction was done using Google Earth Engine (GEE). Considering the accuracy and feasibility of Landsat 8 TIRS, which contain two thermal bands (B10 and B11), a single-channel algorithm was utilized [52,53] as per the USGS's advice against utilizing the split-window for acquiring the LST because of the uncertainties associated with the observations of the second thermal band [15]. Moreover, the 15-min time lag between TM and TIRS is not an obstacle to the analysis. Estimating the LST from the Landsat series necessitates several steps, which include the transformation of digital number (DN) to spectral radiance ($L\lambda$), the calculation of brightness temperature (BT), and adjusting the emissivity [54–56].

The spectral radiance of Landsat 5 TM band 6 was calculated with the following equation:

$$L\lambda = \left(\frac{LMAX\lambda - LMIN\lambda}{QCALMAX - QCALMIN} \right) * (QCAL - QCALMIN) + LMIN\lambda \quad (1)$$

where $L\lambda$ = Spectral radiance ($Wm^{-2} sr^{-1} \mu m^{-1}$), $QCAL$ = Quantized calibrated pixel value in Digital Number (DN), and $LMIN\lambda$ and $LMAX\lambda$ = Spectral radiance scaled to $QCALMIN$ ($Wm^{-2} sr^{-1} \mu m^{-1}$) and $QCALMAX$ ($Wm^{-2} sr^{-1} \mu m^{-1}$), respectively. $QCALMIN$ and $QCALMAX$ = Minimum and Maximum quantized calibrated pixel value in DN corresponding to $LMIN\lambda$ (DN = 0) and $LMAX\lambda$ (DN = 255).

The top-of-atmosphere (TOA) spectral radiance was calculated using Equation (2) in the case of Landsat 8 TIRS:

$$TOA(L\lambda) = ML * QCAL + AL \quad (2)$$

where ML is the multiplicative rescaling factor of the thermal band (B10), $QCAL$ is the Level 1 pixel value in DN, and AL is the additive rescaling factor of the thermal band.

With the help of two thermal constants and $TOA(L\lambda)$, the Brightness Temperature (BT) was calculated from the following equation [57]:

$$BT = \left(K2 / \left(\ln \left(\frac{K1}{TOA(L\lambda)} \right) + 1 \right) \right) - 273.15 \quad (3)$$

where $K1$ and $K2$ are the calibration constants. For Landsat 5 TM, the $K1$ constant = $607.76 \text{ Wm}^{-2} \text{ sr}^{-1} \mu\text{m}^{-1}$ and $K2 = 1260.56 \text{ Wm}^{-2} \text{ sr}^{-1} \mu\text{m}^{-1}$. For Landsat 8 TIRS the $K1 = 774.8853 \text{ Wm}^{-2} \text{ sr}^{-1} \mu\text{m}^{-1}$ and $K2 = 1321.0789 \text{ Wm}^{-2} \text{ sr}^{-1} \mu\text{m}^{-1}$, and $TOA (L\lambda)$ is spectral radiance in $\text{Wm}^{-2} \text{ sr}^{-1} \mu\text{m}^{-1}$.

Normalized Difference Vegetation Index ($NDVI$) was calculated using Equation (4):

$$NDVI = (NIR - RED) / (NIR + RED) \quad (4)$$

where RED is the reflectance of the red band, and NIR is the reflectance of the near-infrared band.

For further processing to obtain the LST, land surface emissivity (ϵ) was required, which was then calculated using the proportion of vegetation (Pv) using the $NDVI$ threshold as proposed by [52] using Equations (5) and (6):

$$Pv = \left(\frac{NDVI - NDVI_{min}}{NDVI_{max} - NDVI_{min}} \right)^2 \quad (5)$$

$$\epsilon = aPv + b \quad (6)$$

where $a = 0.004$ and $b = 0.986$.

The land surface temperature (LST) was calculated using Equation (7):

$$LST = \left(\frac{BT}{1 + \left(0.00115 \times \frac{BT}{1.4388} \right) * \ln(\epsilon)} \right) \quad (7)$$

3.3. LST and Vegetation Persistence Analysis Using the Hurst Exponent

The Hurst exponent remains a less-explored method to investigate the persistence of trends in vegetation and LST dynamics. Rescaled range analysis (R/S analysis) is a statistical technique that utilizes the Hurst exponent, computed from the auto-covariance function, to evaluate consistency and long-range dependencies in time series data. The auto-covariance function quantifies the correlation of a process with itself at varying time intervals. This Hurst exponent, derived from the auto-covariance, serves as a parameter to determine whether the natural phenomena represented by the time series exhibit persistent or anti-persistent behavior [58]. The auto-covariance function's capacity to capture long-term memory effects that diminish exponentially, coupled with a spectral density that tends towards infinity, renders it particularly advantageous for investigating phenomena characterized by long-range dependencies or fractal-like patterns. This analytical approach has found extensive applications across diverse fields, including geology, hydrology, economics, and climatology, enabling researchers to gain insights into the underlying dynamics of complex systems [59]. The Hurst exponent was calculated using the method proposed by [60] following Equations (8)–(12).

This method involves several steps for analyzing a time series $\{\xi(\tau) (\tau = 1, 2, \dots, n)\}$ by dividing it into τ sub-series $x(t)$, where t ranges from 1 to τ . The key steps are as follows:

1. Mean sequence calculation,

$$\langle \xi \rangle_{\tau} = \frac{1}{\tau} \sum_{t=1}^{\tau} x(t), \tau = 1, 2, \dots, n \quad (8)$$

2. Cumulative deviation computation,

$$X(t, \tau) = \sum_{u=1}^t (\xi(u) - \langle \xi \rangle_{\tau}), 1 \leq t \leq \tau \quad (9)$$

3. Range sequence creation,

$$R(\tau) = \max_{1 \leq t \leq \tau} X(t, \tau) - \min_{1 \leq t \leq \tau} X(t, \tau), \tau = 1, 2, \dots, n \quad (10)$$

4. Standard deviation sequence construction,

$$S(\tau) = \left(\frac{1}{\tau} \sum_{t=1}^{\tau} (\xi(t) - \langle \xi \rangle_{\tau})^2 \right)^{1/2}, \tau = 1, 2, \dots, n \quad (11)$$

5. Range rescaling,

$$\frac{R(\tau)}{S(\tau)} = (c\tau)^H \quad (12)$$

The H value ranges from 0 to 1, with 0.5 representing a random walk (drunkard's walk), which is also known as the Brownian series [61]. A value of H ranging from 0.5 to 1 signifies a “persistent trend”, indicating increasing stability of the data as H approaches 1. Conversely, an H value below 0.5 denotes an “anti-persistent trend”, with values closer to 0 indicating instability within the time series [62,63]. We have further reclassified the values into five classes: H value > 0.7 = strongly persistent, H value 0.56 – 0.69 = weakly persistent, H value 0.46 – 0.55 = random, H value 0.31 – 0.45 = weakly anti-persistent, and H values < 0.3 = strongly anti-persistent. The Hurst exponent was applied to preprocessed NDVI and LST imagery using *pracma*, *raster*, *rgdal*, and *ggplot2* libraries in the R 4.0.2 programming language.

3.4. Calculating the Actual Contribution of LCCs to LST

To evaluate the relationship between LST and LCC transition (calculated by employing a land change modeler LCM), this research utilized a less explored technique, the CFLR model, which is an external extension in ArcGIS and can perform raster-to-raster correlation. The equation for the linear regression is as follows:

$$Y = aX + b \quad (13)$$

where Y represents the dependent variable, X stands for the independent variable, a denotes the intercept, and b represents the slope coefficient. Additionally, the zonal statistics (ArcGIS 10.4) were applied to find the LST and NDVI values to calculate each land cover category's temperature and vegetation trends.

4. Results

4.1. Land Cover Changes under BTTP

Figure 2 indicates the LCC patterns of KPK province in the last two decades, with the statistics in Table 1. An apparent surge in artificial surfaces, which have almost doubled in the previous two decades, can be noted, with an evident increase in grassland and forest at the cost of bare land.

Table 1. Land cover changes in sq. kilometers and percentage from 2000 to 2020 in KPK.

Land Cover Class	2000		2020	
	Square Kilometers	Percentage %	Square Kilometers	Percentage %
Cultivated land	28,196.8	27.95	28,072.8	27.82
Forest	15,506.2	15.37	15,897.0	15.76
Grassland	29,081.6	28.82	31,411.6	31.13
Shrubland	1423.1	1.41	1453.3	1.44
Wetland	138.7	0.14	101.6	0.10
Water bodies	613.9	0.61	781.6	0.77
Artificial Surfaces	668.8	0.66	1142.9	1.13
Bare Land	19,630.4	19.46	16,802.8	16.65
Permanent snow and ice	5635.2	5.59	5232.1	5.19

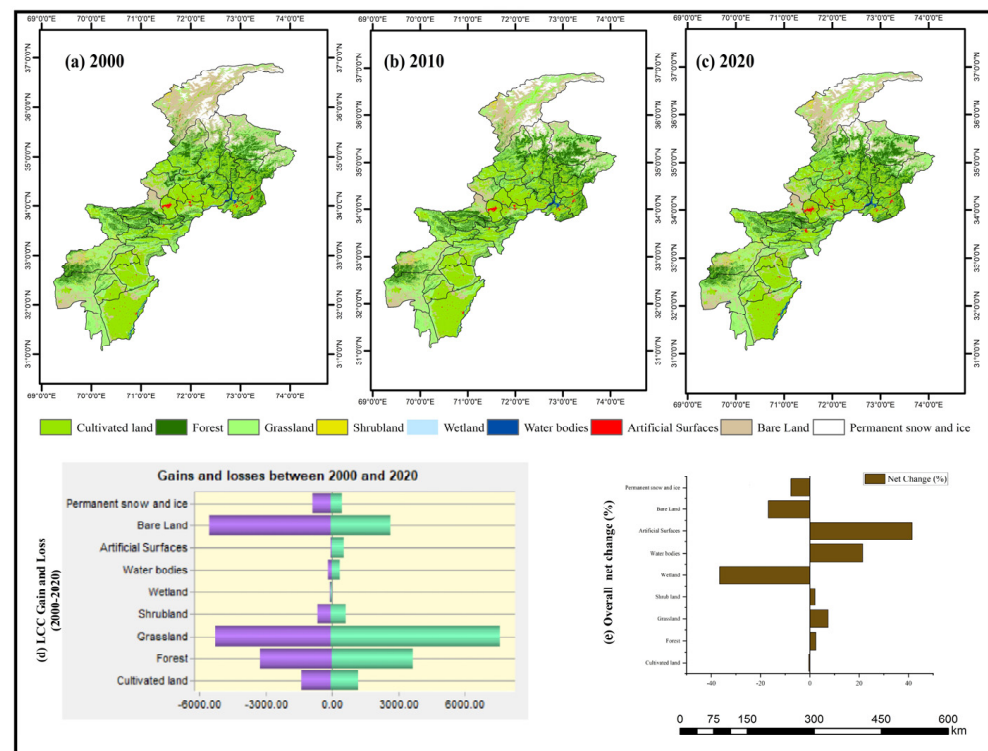


Figure 2. (a–c) Spatial patterns of LCCs in KPK province from 2000–2020, (d) LCC gain and loss in each LC class, (e) net change in each LC class in the last two decades.

The results in Figure 2a–e and Table 1 indicate that the area of artificial surfaces has increased from 688.8 km² (0.66%) in 2000 to 1142.9 km² (1.13%) in 2020. Although a positive growth of green fraction classes (e.g., grassland, forest, shrubland) was observed, the most significant growth was observed in grassland (from 29,081.6 km² (28.82%) in 2000 to 31,411.6 km² (31.13%) in 2020). On the other hand, forest cover has slightly increased, rising from 15,506.20 km² (15.37%) to 15,897.0 km² (15.76%) during the last two decades; contrary to the claims of the government of KPK that the BTTP increased forest by 2% (https://few.kp.gov.pk/page/about_billion_tree_tsunami_afforestation_project (accessed on 24 November 2023)) [43], results from our investigation indicate that forest has increased by merely 0.39%. Although the overall vegetation cover has increased, a significant increase of more than 2% is found in grassland, not in forest cover, which was the prime objective of the BTTP. Similarly, water bodies and shrubland have also revealed an increase from 0.61% to 0.77% and from 1.41% to 1.44%, respectively, of the total land area of KPK from 2000 to 2020. Additionally, a declining trend of cultivated land, permanent snow and ice, and bare land, with reductions from 28,196.8 km² (27.95%), 5635.2 km² (5.59%), and 19,630.4 km² (19.46%) to 28,072.8 km² (27.82%), 5232.1 km² (5.19%), and 16,802.8 km² (16.65%), respectively, was observed.

4.2. Land Cover Change Transitions

After carefully evaluating the LCC trends, we further employed the LCM model to validate the trends and observe the transitions of LCCs to better recognize the interactions between land cover types in the region in the last two decades and specifically after the BTTP. Results in Figure 3 and Table 2 clearly illustrate each LCC transition's spatial locations and area for the last twenty years in the region. Even though an overall change from bare land to vegetation is observed, this does not prove that the objectives of the BTTP were achieved. The transition matrices revealed that an area of 3904.87 km² of bare land has changed into grassland (which was not a prime objective of the BTTP), whereas an area of merely 708.10 km² has changed into forest cover. Meanwhile, an area of 330.30 km² of

forest cover transformed back into bare land, making a net change of just 377.80 km² from bare land to forest, which cannot be considered significant compared to the transformation from bare land to grassland.

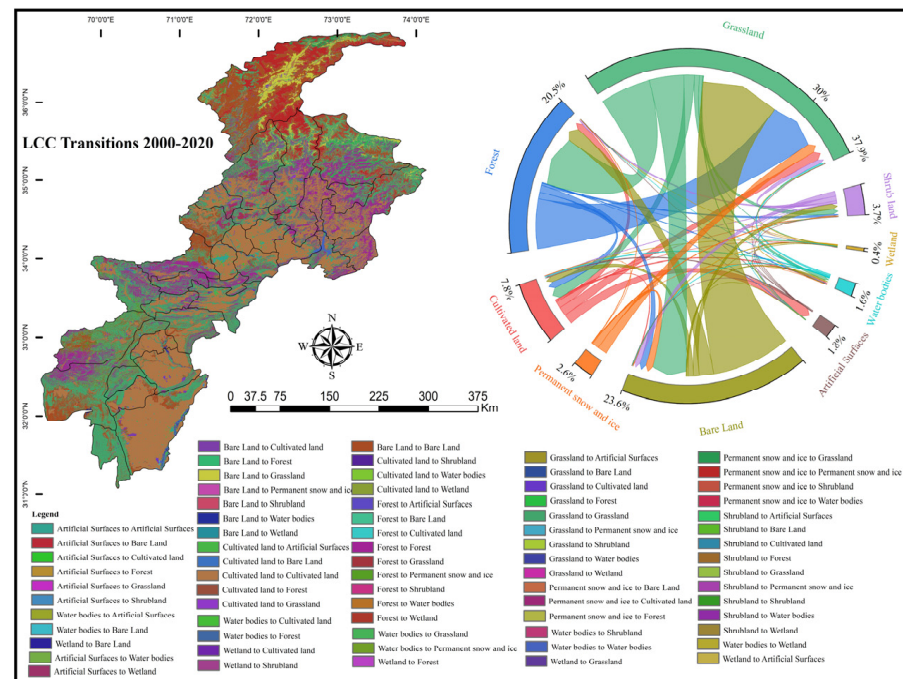


Figure 3. Land Cover Change Transitions from 2000 to 2020 in KPK.

Table 2. Land cover change transitions in square kilometers from 2000 to 2020 in KPK. The table shows the land cover changes from each land cover class in the first column to the land cover classes in the first row in square kilometers.

	Cultivated Land	Forest	Grassland	Shrubland	Wetland	Water Bodies	Artificial Surfaces	Bare Land	Permanent Snow and Ice
Cultivated Land	26,823.2	333.5	441.8	4.4	3.7	68.6	486.8	32.7	
Forest	330.6	12,238.4	2473.2	92.1	3.3	29.9	4.2	330.3	4.1
Grassland	512.7	2493.5	23,780.0	246.7	12.1	154.0	36.5	1702.4	143.8
Shrubland	20.3	92.8	232.4	786.3	0.3	4.2	0.4	246.7	39.8
Wetland	11.7	8.1	12.1	0.1	54.4	23.3	0.2	28.9	
Water bodies	47.7	4.2	86.9	1.3	17.2	426.8	1.3	27.1	1.3
Artificial Surfaces	7.4	3.6	3.3	0.0	0.0	0.7	603.7	0.5	
Bare Land	269.4	708.1	3904.9	277.3	10.6	73.3	7.7	14,093.3	286.0
Permanent Snow and ice	0.0	14.6	477.0	45.0		0.4		340.9	4757.2

In addition, an area of 512.70 km² of grassland, 330.6 km² of forest, and 269.4 km² of bare land have changed into cultivated land, and an area of 486.80 km² of cultivated land has transformed into artificial surfaces in the region over the years considered. Permanent snow and ice have shown a decline in the total area over the years because of transformation into grassland and bare land, with an area of 477.0 km² and 340.9 km², respectively.

4.3. Gain and Loss among LC Classes over the Period 2000 to 2020

After evaluating the LCC trends and transitions, we have applied a land change modeler for ecological sustainability to accurately determine the spatial locations where the LCC transformations have occurred.

Figure 4a shows that most cultivated land remained unaltered with a few losses near the central upper region of the province, where artificial surfaces have shown gains (Table 2). Furthermore, Figure 4b shows that forest cover losses coincide with the grassland class's spatial gain locations. At the same time, losses in grassland spatially coincide with forest cover gains (Figure 4c). The results in Figure 4c,d indicate that the spatial locations of the gains in grassland align with the spatial location of the losses in bare land. The substantial losses in bare land (2828.4 km^2) (Table 1) have contributed a significant portion of the grassland gains (approximately 2300 km^2) (Table 1). Lastly, the losses in permanent snow and ice (around 717 km^2) in Figure 4f are spatially consistent with gains in grassland (Figure 4c) and bare land (Figure 4d). These results also affirm and validate the results from LCC transitions over the study's temporal span in the KPK province.

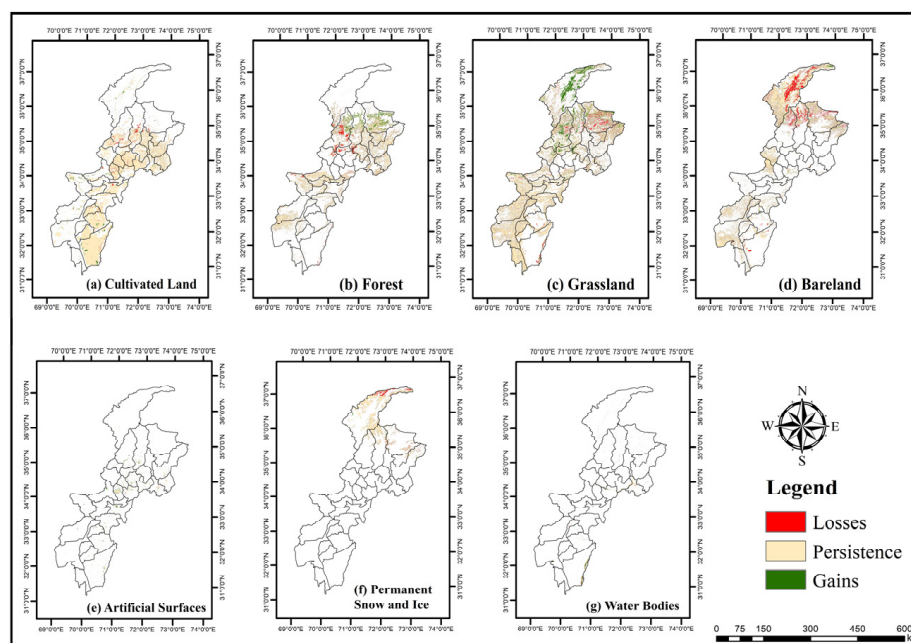


Figure 4. Gain and loss in LC classes in the KPK from 2000 to 2020. (a) Cultivated land, (b) forest, (c) grassland, (d) bare land, (e) artificial surfaces, (f) permanent snow and ice, and (g) water bodies.

4.4. LST Fluctuations 2000–2020

The spatial analysis of LST across three distinct temporal periods (2000, 2010, and 2020) indicated a discernible decline in LST within the region over the past two decades (Figure 5).

The results explicitly revealed an increase in mean LST from 2000 to 2010, with a rise of 1.1°C in the summer and 0.51°C in winter. On the other hand, LST results in Table 3 show that the minimum and maximum LST declined significantly, with a mean decline of 2.60°C in the summer and 2.23°C in the winter season during 2010–2020. Remarkably, this decreasing trend in temperature from 2010 to 2020 coincides with the implementation of the BTTP (2013–2017), signifying a potential correlation between land cover changes induced by the BTTP and the observed LST fluctuations. Furthermore, the minimum LST for the summer and winter seasons was observed in 2020, at -5.68°C and -25.60°C , respectively. The highest values of LST for both seasons were observed in 2010, with 47.94°C for the summer season and 24.42°C for the winter season. Nevertheless, prior investigations have noted similar fluctuations in the KPK region over recent decades. The LST findings of this study are consistent with the most recent research on the subject in the region [39,64,65].

Likewise, significant fluctuations in LST, characterized by increases or decreases, were prevalent in areas exhibiting changes in green fraction.

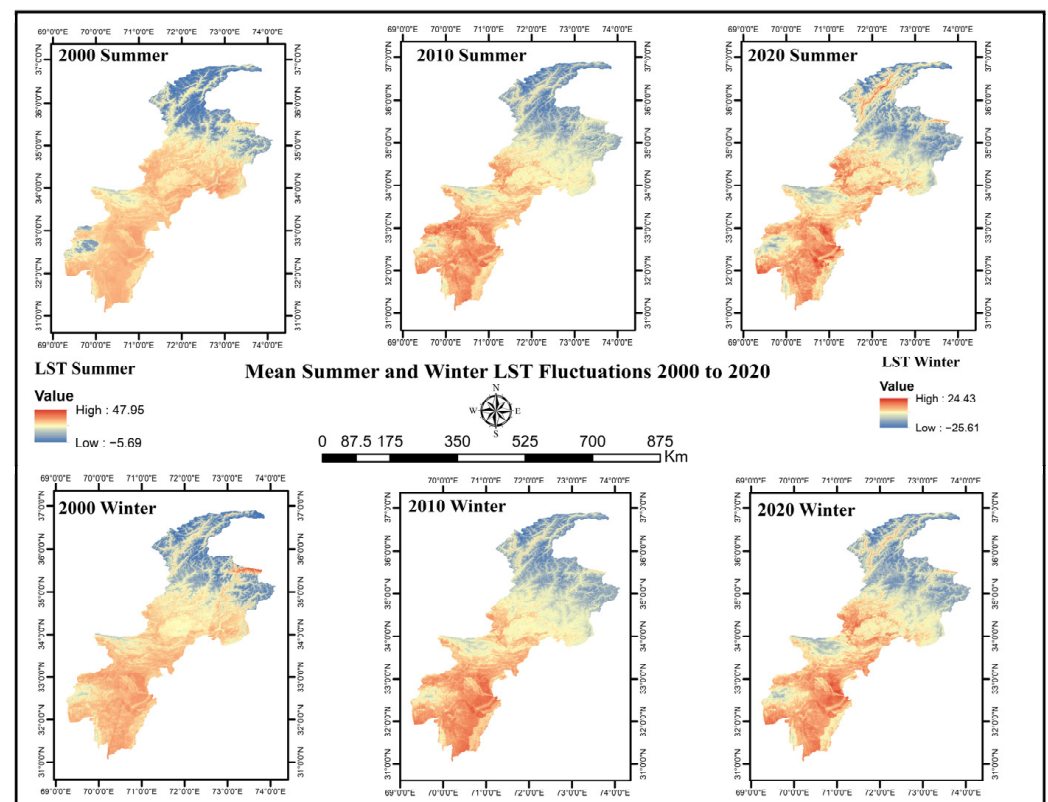


Figure 5. Land surface temperature fluctuations from 2000 to 2020.

Table 3. LST statistics in each study year.

	Summer			Winter		
Years	2000	2010	2020	2000	2010	2020
LST_min (°C)	−3.81	−3.26	−5.68	−24.61	−23.19	−25.60
LST_mean (°C)	30.02	31.12	28.52	10.47	10.98	8.75
LST_max (°C)	47.39	47.94	45.16	23.88	24.43	22.39

4.5. LCC Contribution to LST

Although the observed changes in LST overlap the LCCs induced by the BTTP, all these fluctuations cannot be solely associated with it, as several other climatic and anthropogenic factors could also be involved. To avoid these concerns, we have applied the CFLR model along with zonal statistics to calculate the actual contributions of each land cover category towards LST (Figure 6). The results vividly elaborated the robust correlation between LST and LCCs in summer and winter. A robust positive correlation was identified between LST and artificial surfaces, and LST and bare land, with correlation coefficients (R values) ranging from 0.46 to 0.69. Conversely, pixels containing water bodies and forest cover exhibited a robust negative correlation, with R values ranging from −0.52 to −0.78 (Figure 6).

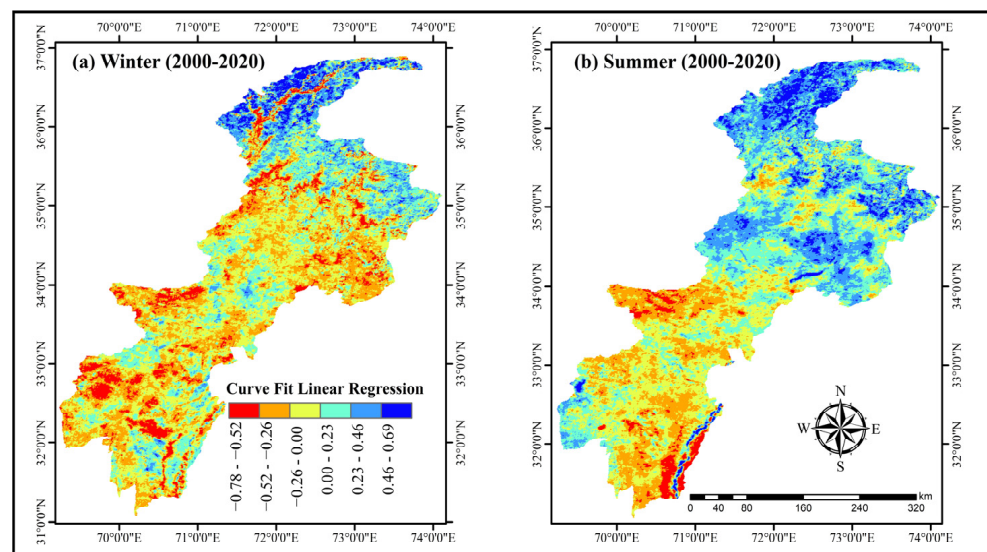


Figure 6. CRLF between LST and LCCs from 2000–2020, (a) winter and (b) summer.

Figure 7a explicitly shows that the land cover changed from wetland to bare land, and artificial surfaces contributed to the highest increase in LST, $0.86\text{ }^{\circ}\text{C}$ and $0.68\text{ }^{\circ}\text{C}$, respectively, in the summer. The transition from grassland to artificial surfaces contributed the most to the LST, with an increase of $0.95\text{ }^{\circ}\text{C}$ (Figure 7b) in the winter season from 2000 to 2020 in the region. Furthermore, all the vegetation classes, i.e., cultivated land, forest, grassland, and shrubland, contributed to the increase in LT when these classes changed into bare land and artificial surfaces. The highest decrease in LST was found in land cover change from artificial surfaces to forest, $-0.85\text{ }^{\circ}\text{C}$ to be precise, in the summer season (Figure 7a). In contrast, the most significant decrease in LST during the winter season was observed in land cover transitions from forest, grassland, shrubland, and bare land to permanent snow and ice with $-0.86\text{ }^{\circ}\text{C}$, $-0.74\text{ }^{\circ}\text{C}$, $-0.86\text{ }^{\circ}\text{C}$, and $-0.73\text{ }^{\circ}\text{C}$, respectively (Figure 7b). These analyses and results from the negative temperature contributions of vegetation classes to LST have validated the previous observations of LST decrease after the implementation of BTTP in the region.

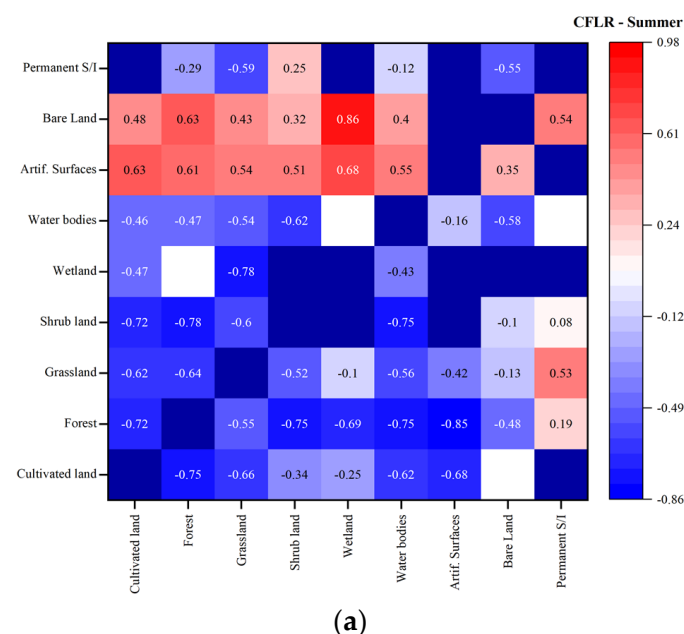


Figure 7. Cont.

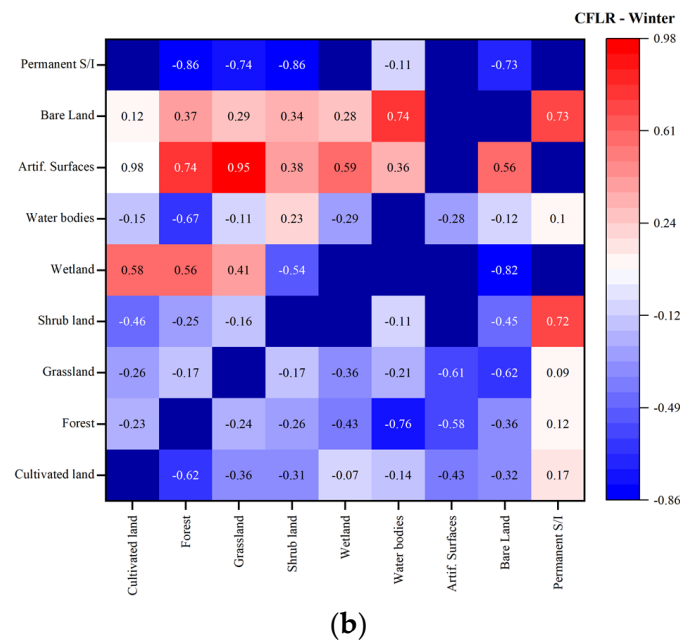


Figure 7. Contributions of each land cover change to the LST in summer (a) and winter (b). (The land cover classes along the x-axis have changed into the land cover classes on the y-axis).

4.6. LST and NDVI Trends Persistence in the Region

The results of linear trend analysis, conducted on NDVI-min, NDVI-max, and NDVI-mean metrics to assess vegetation trends within the region, utilizing mean annual growing season (May–Nov) NDVI data from 2000 to 2020, derived from Landsat 5 and Landsat 8 imagery processed through Google Earth Engine, show an increasing trend (Figure 8b). As illustrated in (Figure 8b,c), apparent upward trends were observed across NDVI-min, NDVI-max, and NDVI-mean values. Preceding the initiation of the BTTP in 2013, the linear trend exhibited irregular variations. However, from 2014 to 2020, a consistent and noticeable upward trajectory is evident, revealing positive vegetation growth trends within the province. To further check the persistence of these trends, we have applied the Hurst exponent using the R programming language; the results are presented in Figure 8a. The Hurst value ranged between 0.22 and 0.74 and was classified into five categories. It is apparent from Figure 8a that a significant portion of the province has fallen under the weakly persistent category, which ultimately signaled that the vegetation growth trend may not last long. The LCC results revealed the reason for this weakly persistent vegetation trend, as the substantial transition under BTTP was to grassland instead of forest (Table 1). The areas with a strongly persistent trend correspond to areas of maximum forest cover.

As demonstrated by the linear trend analysis of mean LST in Figure 9c,d, the mean LST has shown an upsurge in the first decade (2000–2010) both in summer and winter, whereas it shows a substantial reduction in the second decade (2010–2020). To further check the persistence of these LST trends, we again applied the Hurst exponent to the mean summer and winter LST imagery of 2000, 2010, and 2020 (Figure 9). The Hurst value ranged between 0.21 and 0.71 for the summer, and 0.24 and 0.78 for the winter. A large proportion of the province demonstrated a weakly persistent trend for LST (Figure 9a,b), validating the results from LCCs and Figure 8a at the same time. Furthermore, the LST trends are more likely to remain strongly persistent (keep decreasing) in the winter season ($H = 0.78$) as compared to the summer season ($H = 0.71$).

The upper part of the province is mainly a snow-covered region with negligible changes in the LST over the year, and hence showed a strongly non-persistent trend. Additionally, the spatial locations of the weakly persistent trends in vegetation and LST

correspond to the spatial locations where the grassland gains occurred (Figure 4c) after the BTTP.

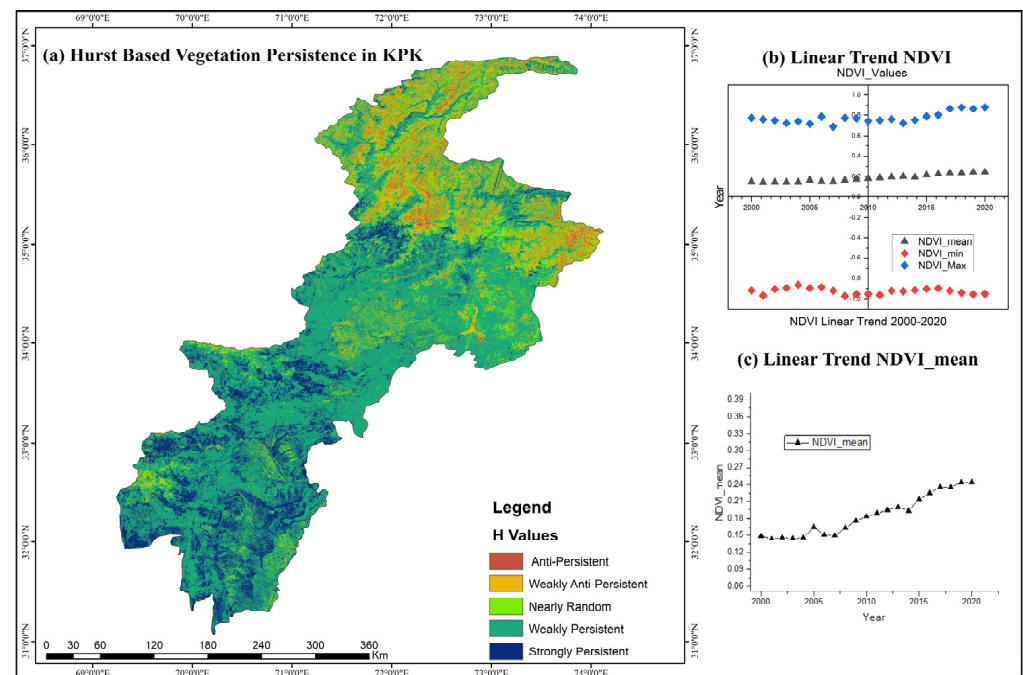


Figure 8. (a) Hurst-based vegetation persistence in KPK. (b) Linear trend analysis NDVI-min, NDVI-max, and NDVI-mean (c) Linear Trend analysis NDVI-mean.

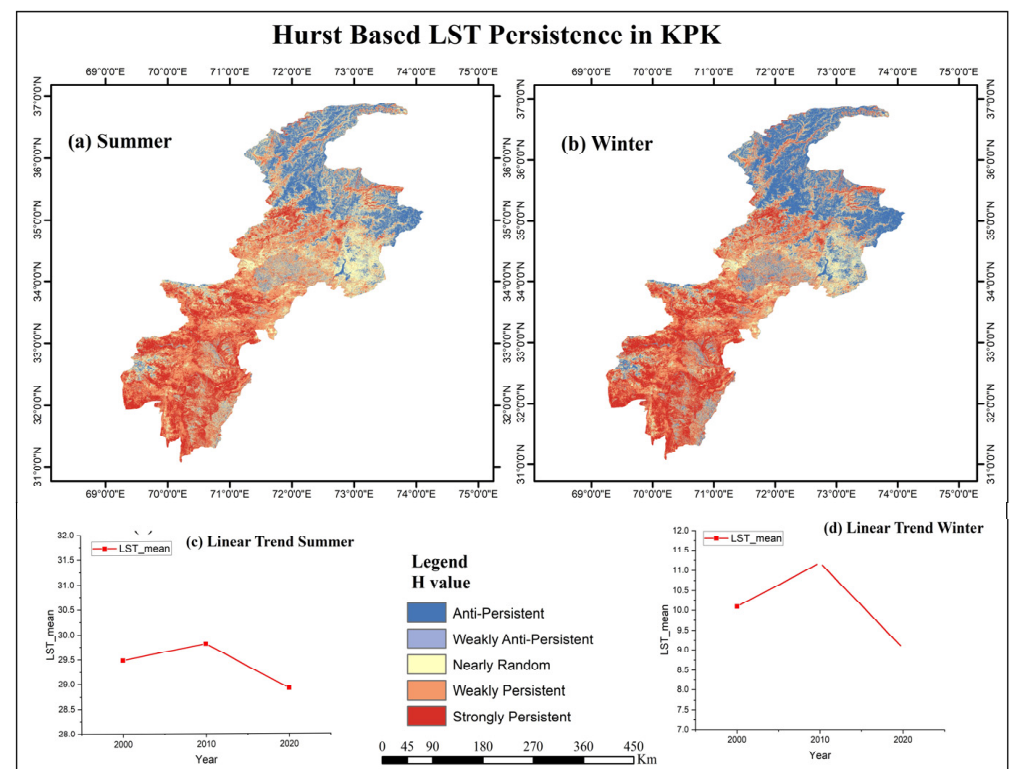


Figure 9. (a,b) Hurst-based LST persistence in KPK. (c) Linear Trend Analysis of mean LST in summer, (d) Linear Trend Analysis of mean LST in winter.

5. Discussion

Global climate has been changing swiftly: land cover changes have increased by four times in the last four decades [66]. Pakistan has been ranked fifth among countries most susceptible to the severe impacts of climate change [39], including temperature increases, flooding, and desertification [67,68]. The mean temperature of Pakistan has already increased by 0.62 °C, and it is expected to increase by 1.2 °C to 6.5 °C by the end of this century [13,69,70]. Different afforestation projects have been implemented globally to fight climate change, and BTTP is one of those projects aimed at restoring forest cover and degraded land.

This research was carried out to investigate the LCCs induced by the BTTP, an afforestation project initiated by the provincial government, and their impact on LST in the KPK province of Pakistan from 2000 to 2020. Furthermore, vegetation types, trends, and persistence were examined in detail. The LCM provided insights into LC class transitions, particularly after the implementation of BTTP in the second decade of this temporal period, revealing the spatial locations and areas of each transition. The findings revealed the challenges in achieving the prime objective of the BTTP, as the most prominent increase was in grassland (increased from 29,081.6 km² (28.82%) in 2000 to 31,411.6 km² (31.13%), more than 2%), which raises questions about the effectiveness of the afforestation project. Furthermore, cultivated land, bare land, and permanent snow and ice have experienced a decrease in the past 20 years.

Additionally, a significant increase in artificial surfaces over the region in the last two decades has been observed, increasing from 668 km² in 2000 to 1143 km² in 2020. A significant part of this increase is associated with the human population surge, from 17.7 million (1998) to 40.85 million (2023), which was partly contributed by mass migration from Afghanistan into the region [71]. Moreover, the findings indicated that the expansion of artificial surfaces comes at the expense of vegetation cover, a factor crucial for maintaining equilibrium among land surface and atmospheric parameters [72–74]. Recent studies [16,72,75] have argued that transition in LC classes, specifically an extension of artificial surfaces, affects the LST and severely influences the local and regional climate. The diversity in surface roughness and reflectance among different land cover classes leads to corresponding variations in the absorption and radiation of energy [76–78].

This study was carried out to explore these arguments by assessing the spatiotemporal fluctuations in LST associated with each LC category. The results of the current study revealed a significant decrease in mean seasonal LST from 2010 to 2020, with a mean decline of 2.60 °C in the summer and 2.23 °C in the winter. To further explore the relationship between LST and vegetation cover and the contributions of each LC class to LST, the current study utilized the CFLR model and zonal statistics [13,40,79,80], and the investigations revealed a strong negative correlation between vegetation classes and LST. Additionally, the results show a significant increase in LST ranging between 0.51 °C and 0.93 °C for both summer and winter seasons in response to transforming any vegetation class into an artificial surface. On the other hand, there was a significant decrease in LST in response to any artificial surface or bare land changing into any vegetation category, ranging between −0.39 °C and −0.81 °C.

A notable decline in LST has been found in linear trend analysis after the BTTP project, indicating the importance of green cover proliferation. Several studies have also highlighted the importance of vegetation cover regarding this occurrence; e.g., [59] reported that protection of biodiversity and green cover, encompassing forests and agricultural land, as well as dense or sparse vegetation, is crucial for preserving soil integrity and ecosystem equilibrium. This is due to the varying surface reflectance and roughness associated with each land cover category, which collectively influence the hydrological cycle, surface temperatures, and atmospheric conditions [14,81]. Recent studies [64,65,82] have discussed the role of BTTP in LST decline in the region; hence, the results of this study were in line with the literature, and the application of the Hurst exponent to both LST and vegetation cover has demonstrated the temporal persistence of the trends [60,83–86]. The Hurst

exponent is commonly utilized to identify the stability in time series data and provide more precise outcomes than other trend analyses. The findings from the Hurst exponent revealed that a significant portion of the province has shown a weakly persistent trend of vegetation, especially in the central region of the KPK, where the most-increased class is grassland. The weak persistence detected in both LST and vegetation indicates that the effects of land cover changes, particularly the increase in grassland, may not exhibit long-term stability in the region.

6. Conclusions

This study aimed to evaluate the LCCs under the BTTP and their influence on LST in the KPK region from 2000 to 2020. The research results reveal that the BTTP afforestation initiative fell short of its primary objective of augmenting the forest cover in the region by a minimum of 2%. Instead, a substantial change to grassland from bare land was observed (3904.87 km²), indicating a deviation from the intended outcome. Further, those LCC transitions have caused a decline of 2.60 °C in the summer and 2.23 °C in LST in the winter season from 2010 to 2020, indicating that this project was not an overall failure. The CFLRM-based results indicate a strong negative correlation between LST and vegetation classes, with an R-value up to −0.78, and a strong positive correlation, with an R-value up to 0.69, between artificial surfaces and LST. The conversion from artificial surfaces and bare land to forest cover, grassland, and shrubland has markedly reduced the mean seasonal LST. Conversely, changes from each vegetation class to bare land and artificial surfaces have contributed to an escalation in mean seasonal LST in the region from 2000 to 2020. The Hurst exponent analysis indicated that a substantial proportion of the study area is characterized by weakly persistent behavior, signifying that the increasing vegetation and decreasing LST trend will not continue for the long term.

Given these insights, future afforestation projects should prioritize increasing forest cover to mitigate the adverse impacts of land degradation and facilitate restoration. Authorities should adjust their strategies instead of replicating the BTTP, as the project achieved a mere 0.39% increase in forest cover. While the project has succeeded in temporarily reducing LST, its long-term benefits are questionable, especially considering that the predominant vegetation cover increase resulting from the initiative is grassland. Consequently, a more nuanced and targeted approach is warranted to achieve sustainable and meaningful afforestation outcomes.

Author Contributions: Conceptualization, H.U.H., H.L. and Q.L.; data curation, H.U.H. and H.L.; formal analysis, H.U.H., H.L. and B.B.; funding acquisition, H.L. and Q.L.; investigation, H.U.H. and H.L.; methodology, H.U.H., H.L. and S.Z.; project administration, H.L. and Q.L.; resources, H.L. and Q.L.; software H.U.H.; supervision, H.L. and Q.L.; validation, H.U.H., H.L., Q.L., B.B., T.H. and S.Z.; visualization, H.U.H.; writing—original draft, H.U.H.; writing—review, and editing, H.U.H., H.L., Q.L., B.B., T.H. and S.Z. All authors have read and agreed to the published version of the manuscript.

Funding: This work was supported by the Chinese Natural Science Foundation Project (41930111, 42071317, 42271362, and 42130111).

Institutional Review Board Statement: Not Applicable.

Informed Consent Statement: Not Applicable.

Data Availability Statement: All the raw satellite imageries used in this study are freely available on the Google Earth Engine's data base (<https://developers.google.com/earth-engine/datasets/>) (accessed on 20 February 2024). Land cover data sources are provided in the methods sections. Further, the ready to use data supporting the conclusions of this article will be made available by the authors on request.

Acknowledgments: Authors Hammad Ul Hussan, Hua Li, Qinhua LIU, Barjeece Bashir, Tian Hu, and Shouyi Zhong acknowledge the Aerospace Information Research Institute, Chinese Academy of Sciences (AIRCAS), University of Chinese Academy of Sciences (UCAS), Beijing, China, and Alliance of International Science Organizations (ANSO).

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Wang, H.; Liu, Y.; Wang, Y.; Yao, Y.; Wang, C. Land cover change in global drylands: A review. *Sci. Environ.* **2023**, *863*, 160943. [\[CrossRef\]](#)
2. UNPF. An urbanizing world. In *Urbanization*, 03 OCT 2016 ed.; United Nations Population Fund: Online, 2016.
3. Naeem, S.; Cao, C.; Waqar, M.M.; Wei, C.; Acharya, B.K. Vegetation role in controlling the ecoenvironmental conditions for sustainable urban environments: A comparison of Beijing and Islamabad. *J. Appl. Remote Sens.* **2018**, *12*, 016013. [\[CrossRef\]](#)
4. Xing, Z.; Li, Z.-L.; Duan, S.-B.; Liu, X.; Zheng, X.; Leng, P.; Gao, M.; Zhang, X.; Shang, G. Estimation of daily mean land surface temperature at global scale using pairs of daytime and nighttime MODIS instantaneous observations. *ISPRS J. Photogramm. Remote Sens.* **2021**, *178*, 51–67. [\[CrossRef\]](#)
5. Yu, P.; Zhao, T.; Shi, J.; Ran, Y.; Jia, L.; Ji, D.; Xue, H. Global spatiotemporally continuous MODIS land surface temperature dataset. *Sci. Data* **2022**, *9*, 143. [\[CrossRef\]](#)
6. Li, H.; Li, R.; Yang, Y.; Cao, B.; Bian, Z.; Hu, T.; Du, Y.; Sun, L.; Liu, Q. Temperature-based and radiance-based validation of the collection 6 MYD11 and MYD21 land surface temperature products over barren surfaces in northwestern China. *IEEE Trans. Geosci. Remote Sens.* **2020**, *59*, 1794–1807. [\[CrossRef\]](#)
7. Njoku, E.A.; Tenenbaum, D.E. Quantitative assessment of the relationship between land use/land cover (LULC), topographic elevation and land surface temperature (LST) in Ilorin, Nigeria. *Remote Sens. Appl. Soc. Environ.* **2022**, *27*, 100780. [\[CrossRef\]](#)
8. Fang, Z.; Ding, T.; Chen, J.; Xue, S.; Zhou, Q.; Wang, Y.; Wang, Y.; Huang, Z.; Yang, S. Impacts of land use/land cover changes on ecosystem services in ecologically fragile regions. *Sci. Total Environ.* **2022**, *831*, 154967. [\[CrossRef\]](#) [\[PubMed\]](#)
9. Xu, X.; Jiang, H.; Wang, L.; Guan, M.; Zhang, T.; Qiao, S. Major consequences of land-use changes for ecosystems in the future in the agro-pastoral transitional zone of northern China. *Appl. Sci.* **2020**, *10*, 6714. [\[CrossRef\]](#)
10. Rajagopal, P.; Priya, R.S.; Senthil, R. A review of recent developments in the impact of environmental measures on urban heat island. *Sustain. Cities Soc.* **2023**, *88*, 104279. [\[CrossRef\]](#)
11. Smith, R.; Choudhury, B.J. On the correlation of indices of vegetation and surface temperature over south-eastern Australia. *Int. J. Remote Sens.* **1990**, *11*, 2113–2120. [\[CrossRef\]](#)
12. Hope, A.S.; McDowell, T. The relationship between surface temperature and a spectral vegetation index of a tallgrass prairie: Effects of burning and other landscape controls. *Int. J. Remote Sens.* **1992**, *13*, 2849–2863. [\[CrossRef\]](#)
13. Mumtaz, F.; Tao, Y.; de Leeuw, G.; Zhao, L.; Fan, C.; Elnashar, A.; Bashir, B.; Wang, G.; Li, L.; Naeem, S. Modeling spatio-temporal land transformation and its associated impacts on land surface temperature (LST). *Remote Sens.* **2020**, *12*, 2987. [\[CrossRef\]](#)
14. Hou, G.; Zhang, H.; Wang, Y.; Qiao, Z.; Zhang, Z. Retrieval and spatial distribution of land surface temperature in the middle part of Jilin province based on MODIS data. *Sci. Geogr. Sin.* **2010**, *30*, 421–427.
15. Zhang, F.; Tiyyip, T.; Kung, H.; Johnson, V.C.; Maimaitiyiming, M.; Zhou, M.; Wang, J. Dynamics of land surface temperature (LST) in response to land use and land cover (LULC) changes in the Weigan and Kuqa river oasis, Xinjiang, China. *Arab. J. Geosci.* **2016**, *9*, 499. [\[CrossRef\]](#)
16. Fatemi, M.; Narangifard, M. Monitoring LULC changes and its impact on the LST and NDVI in District 1 of Shiraz City. *Arab. J. Geosci.* **2019**, *12*, 127. [\[CrossRef\]](#)
17. Hu, Y.; Raza, A.; Syed, N.R.; Acharki, S.; Ray, R.L.; Hussain, S.; Dehghanisanij, H.; Zubair, M.; Elbeltagi, A. Land Use/Land Cover Change Detection and NDVI Estimation in Pakistan's Southern Punjab Province. *Sustainability* **2023**, *15*, 3572. [\[CrossRef\]](#)
18. Li, H.; Li, R.; Tu, H.; Cao, B.; Liu, F.; Bian, Z.; Hu, T.; Du, Y.; Sun, L.; Liu, Q. An Operational Split-Window Algorithm for Generating Long-Term Land Surface Temperature Products from Chinese Fengyun-3 Series Satellite Data. *IEEE Trans. Geosci. Remote Sens.* **2023**, *61*. [\[CrossRef\]](#)
19. Li, R.; Li, H.; Hu, T.; Bian, Z.; Liu, F.; Cao, B.; Du, Y.; Sun, L.; Liu, Q. Land Surface Temperature Retrieval from Sentinel-3A SLSTR data: Comparison among Split-window, Dual-window, Three-channel and Dual-angle algorithms. *IEEE Trans. Geosci. Remote Sens.* **2023**, *61*. [\[CrossRef\]](#)
20. Krivoguz, D.; Chernyi, S.G.; Zinchenko, E.; Silkin, A.; Zinchenko, A. Using Landsat-5 for accurate historical LULC classification: A comparison of machine learning models. *Data* **2023**, *8*, 138. [\[CrossRef\]](#)
21. Rehman, Z.; Kazmi, S.J.H.; Khanum, F.; Samoon, Z.A. Analysis of land surface temperature and NDVI using geo-spatial technique: A case study of Keti Bunder, Sindh, Pakistan. *J. Basic Appl. Sci.* **2015**, *11*, 514–527. [\[CrossRef\]](#)
22. Hussain, M.; Butt, A.R.; Uzma, F.; Ahmed, R.; Irshad, S.; Rehman, A.; Yousaf, B. A comprehensive review of climate change impacts, adaptation, and mitigation on environmental and natural calamities in Pakistan. *Environ. Monit. Assess.* **2020**, *192*, 48. [\[CrossRef\]](#) [\[PubMed\]](#)
23. Irons, J.R.; Dwyer, J.L.; Barsi, J.A. The next Landsat satellite: The Landsat data continuity mission. *Remote Sens. Environ.* **2012**, *122*, 11–21. [\[CrossRef\]](#)
24. Ermida, S.L.; Soares, P.; Mantas, V.; Götsche, F.-M.; Trigo, I.F. Google earth engine open-source code for land surface temperature estimation from the landsat series. *Remote Sens.* **2020**, *12*, 1471. [\[CrossRef\]](#)
25. Schwarz, N.; Lautenbach, S.; Seppelt, R. Exploring indicators for quantifying surface urban heat islands of European cities with MODIS land surface temperatures. *Remote Sens. Environ.* **2011**, *115*, 3175–3186. [\[CrossRef\]](#)

26. Ullah, S.; Ahmad, K.; Sajjad, R.U.; Abbasi, A.M.; Nazeer, A.; Tahir, A.A. Analysis and simulation of land cover changes and their impacts on land surface temperature in a lower Himalayan region. *J. Environ. Manag.* **2019**, *245*, 348–357. [CrossRef] [PubMed]
27. Zhao, W.; He, J.; Wu, Y.; Xiong, D.; Wen, F.; Li, A. An analysis of land surface temperature trends in the central Himalayan region based on MODIS products. *Remote Sens.* **2019**, *11*, 900. [CrossRef]
28. IPCC—Intergovernmental Panel on Climate Change. Climate Change 2021: The Physical Science Basis. Available online: <https://www.ipcc.ch/report/ar6/wg1/> (accessed on 21 December 2023).
29. Xiao, H.; Weng, Q. The impact of land use and land cover changes on land surface temperature in a karst area of China. *J. Environ. Manag.* **2007**, *85*, 245–257. [CrossRef] [PubMed]
30. Hamdi, R. Estimating urban heat island effects on the temperature series of Uccle (Brussels, Belgium) using remote sensing data and a land surface scheme. *Remote Sens.* **2010**, *2*, 2773–2784. [CrossRef]
31. Ovalle, A.; Tristán, A.; Amador-Nieto, J.; Putri, R.; Zahra, R. Analysing the land use/land cover influence on land surface temperature in San Luis Potosí Basin, México using remote sensing techniques. *IOP Conf. Ser. Earth Environ. Sci.* **2021**, *686*, 012029. [CrossRef]
32. Jalan, S.; Sharma, K. Spatio-temporal assessment of land use/land cover dynamics and urban heat island of Jaipur city using satellite data. *Int. Arch. Photogramm. Remote Sens. Spat. Inf. Sci.* **2014**, *40*, 767–772. [CrossRef]
33. Tollefson, J. The 2 C dream. *Nature* **2015**, *527*, 436. [CrossRef]
34. Mansourian, S.; Berrahmouni, N. *Review of forest and landscape restoration in Africa 2021*; Food & Agriculture Organization: Rome, Italy, 2021. [CrossRef]
35. Zhai, J.; Wang, L.; Liu, Y.; Wang, C.; Mao, X. Assessing the effects of China's three-north shelter forest program over 40 years. *Sci. Total Environ.* **2023**, *857*, 159354. [CrossRef]
36. De Pinto, A.; Cenacchi, N.; Robertson, R.; Kwon, H.-Y.; Thomas, T.; Koo, J.; Begeladze, S.; Kumar, C. The role of crop production in the forest landscape restoration approach—Assessing the potential benefits of meeting the Bonn Challenge. *Front. Sustain. Food Syst.* **2020**, *4*, 61. [CrossRef]
37. Fagan, M.E.; Reid, J.L.; Holland, M.B.; Drew, J.G.; Zahawi, R.A. How feasible are global forest restoration commitments? *Conserv. Lett.* **2020**, *13*, e12700. [CrossRef]
38. Pakistan Bureau of Statistics, P. The Population of Pakistan reaches 241.49 Million as the Digital Census Concludes. Available online: <https://www.pbs.gov.pk/sites/default/files/population/2023/Press%20Release.pdf> (accessed on 18 September 2023).
39. Mumtaz, F.; Li, J.; Liu, Q.; Tariq, A.; Arshad, A.; Dong, Y.; Zhao, J.; Bashir, B.; Zhang, H.; Gu, C. Impacts of green fraction changes on surface temperature and carbon emissions: Comparison under forestation and urbanization reshaping scenarios. *Remote Sens.* **2023**, *15*, 859. [CrossRef]
40. Tariq, A.; Jiango, Y.; Li, Q.; Gao, J.; Lu, L.; Soufan, W.; Almutairi, K.F.; Habib-ur-Rahman, M. Modelling, mapping and monitoring of forest cover changes, using support vector machine, kernel logistic regression and naive bayes tree models with optical remote sensing data. *Heliyon* **2023**, *9*, 13212. [CrossRef] [PubMed]
41. Forest Carbon Partnership Facility (FCPF). *Readiness Preparation Proposal (R-PP) for Pakistan*; Forest Carbon Partnership Facility (FCPF): Islamabad, Pakistan, 2013; p. 158.
42. Bhatti, S. An over view of forests in Pakistan. *World Environ. Day. Pakistan.* **2011**, *16*, 88–97.
43. Pakhtunkhwa, G.O.K. Billion Tree Tsunami Afforestation Project. Available online: https://few.kp.gov.pk/page/about_billion_tree_tsunami_afforestation_project (accessed on 24 November 2023).
44. Pakhtunkhwa, G.O.K. Billion Tree Tsunami—A Dream Came True. Available online: <https://www.insaf.pk/news/billion-tree-tsunami-dream-came-true> (accessed on 24 November 2023).
45. Khan, M.D.; Ali, A.; Iqbal, A.; Rehman, Z.-U.; Ahmed, W.; Arooj, S.A. Monitoring of Afforestation Activities Using Landsat-8 Temporal Images, Billion Trees Afforestation Project, Pakistan. *Nova Meh. Šumarstva Časopis Za Teor. I Praksu Šumarskoga Inženjerstva* **2021**, *42*, 27–40.
46. Abbas, S.; Qamer, F.M.; Ali, H.; Usman, M.; Ahmad, A.; Salman, A.; Akhter, A.M. Monitoring of large-scale forest restoration: Evidence of vegetation recovery and reversing chronic ecosystem degradation in the mountain region of Pakistan. *Ecol. Inform.* **2023**, *77*, 102277. [CrossRef]
47. Tariq, A.; Yan, J.; Mumtaz, F. Land change modeler and CA-Markov chain analysis for land use land cover change using satellite data of Peshawar, Pakistan. *Phys. Chem. Earth Parts A/B/C* **2022**, *128*, 103286. [CrossRef]
48. Qazi, M.S.; Faisal, S. Annotated Geological Bibliography of Northern Pakistan up to Ad 2002. 2017, Volume 1, pp. 264–286. Available online: https://www.researchgate.net/publication/324189536_Annotated_Geological_Bibliography_of_northern_Pakistan_up_to_AD_2002 (accessed on 17 March 2024).
49. Wang, M.; He, C.; Zhang, Z.; Hu, T.; Duan, S.-B.; Mallick, K.; Li, H.; Liu, X. Evaluation of Three Land Surface Temperature Products from Landsat Series Using in Situ Measurements. *IEEE Trans. Geosci. Remote Sens.* **2023**, *61*, 1–19. [CrossRef]
50. Eastman, J.R.; Toledano, J. A short presentation of the Land Change Modeler (LCM). In *Lecture Notes in Geoinformation and Cartography*; Springer: Cham, Switzerland, 2018; pp. 499–505.
51. Mishra, V.N.; Rai, P.K.; Mohan, K. Prediction of land use changes based on land change modeler (LCM) using remote sensing: A case study of Muzaffarpur (Bihar), India. *J. Geogr. Inst. Jovan Cvijic SASA* **2014**, *64*, 111–127. [CrossRef]
52. Qacami, M.; Khattabi, A.; Lahssini, S.; Rifai, N.; Meliho, M. Land-cover/land-use change dynamics modeling based on land change modeler. *Ann. Reg. Sci.* **2023**, *70*, 237–258. [CrossRef]

53. Jiménez-Muñoz, J.C.; Sobrino, J.A. A generalized single-channel method for retrieving land surface temperature from remote sensing data. *J. Geophys. Res. Atmos.* **2003**, *108*. [\[CrossRef\]](#)
54. Wang, M.; Zhang, Z.; Hu, T.; Liu, X. A practical single-channel algorithm for land surface temperature retrieval: Application to landsat series data. *J. Geophys. Res. Atmos.* **2019**, *124*, 299–316. [\[CrossRef\]](#)
55. Avdan, U.; Jovanovska, G. Algorithm for automated mapping of land surface temperature using LANDSAT 8 satellite data. *J. Sens.* **2016**, *2016*, 1480307. [\[CrossRef\]](#)
56. Govil, H.; Guha, S.; Diwan, P.; Gill, N.; Dey, A. Analyzing linear relationships of LST with NDVI and MNDISI using various resolution levels of Landsat 8 OLI and TIRS data. In *Data Management, Analytics and Innovation. Advances in Intelligent Systems and Computing*; Springer: Singapore, 2020; Volume 1, pp. 171–184.
57. Tariq, A.; Shu, H. CA-Markov chain analysis of seasonal land surface temperature and land use land cover change using optical multi-temporal satellite data of Faisalabad, Pakistan. *Remote Sens.* **2020**, *12*, 3402. [\[CrossRef\]](#)
58. Ayanlade, A.; Aigbiremolen, M.I.; Oladosu, O.R. Variations in urban land surface temperature intensity over four cities in different ecological zones. *Sci. Rep.* **2021**, *11*, 20537. [\[CrossRef\]](#)
59. Emamian, A.; Rashki, A.; Kaskaoutis, D.G.; Gholami, A.; Opp, C.; Middleton, N. Assessing vegetation restoration potential under different land uses and climatic classes in northeast Iran. *Ecol. Indic.* **2021**, *122*, 107325. [\[CrossRef\]](#)
60. Bashir, B.; Cao, C.; Naeem, S.; Zamani Joharestani, M.; Bo, X.; Afzal, H.; Jamal, K.; Mumtaz, F. Spatio-temporal vegetation dynamic and persistence under climatic and anthropogenic factors. *Remote Sens.* **2020**, *12*, 2612. [\[CrossRef\]](#)
61. Peng, J.; Liu, Z.; Liu, Y.; Wu, J.; Han, Y. Trend analysis of vegetation dynamics in Qinghai–Tibet Plateau using Hurst Exponent. *Ecol. Indic.* **2012**, *14*, 28–39. [\[CrossRef\]](#)
62. von Wegner, F.; Tagliazucchi, E.; Brodbeck, V.; Laufs, H. Analytical and empirical fluctuation functions of the EEG microstate random walk-short-range vs. long-range correlations. *Neuroimage* **2016**, *141*, 442–451. [\[CrossRef\]](#) [\[PubMed\]](#)
63. Ahmad, A.; Zhang, J.; Bashir, B.; Mahmood, K.; Mumtaz, F. Exploring vegetation trends and restoration possibilities in Pakistan by using Hurst exponent. *Environ. Sci. Pollut. Res.* **2023**, *30*, 91915–91928. [\[CrossRef\]](#) [\[PubMed\]](#)
64. Appleton, K.; Lovett, A. GIS-based visualisation of rural landscapes: Defining ‘sufficient’ realism for environmental decision-making. *Landsc. Urban Plan.* **2003**, *65*, 117–131. [\[CrossRef\]](#)
65. Khan, R.; Li, H.; Basir, M.; Chen, Y.L.; Sajjad, M.M.; Haq, I.U.; Ullah, B.; Arif, M.; Hassan, W. Monitoring land use land cover changes and its impacts on land surface temperature over Mardan and Charsadda Districts, Khyber Pakhtunkhwa (KP), Pakistan. *Environ. Monit. Assess.* **2022**, *194*, 409. [\[CrossRef\]](#) [\[PubMed\]](#)
66. Nasir, M.J.; Ahmad, W.; Iqbal, J.; Ahmad, B.; Abdo, H.G.; Hamdi, R.; Batani, S.M. Effect of the urban land use dynamics on land surface temperature: A case study of kohat city in Pakistan for the period 1998–2018. *Earth Syst. Environ.* **2022**, *6*, 237–248. [\[CrossRef\]](#)
67. Winkler, K.; Fuchs, R.; Rounsevell, M.; Herold, M. Global land use changes are four times greater than previously estimated. *Nat. Commun.* **2021**, *12*, 2501. [\[CrossRef\]](#) [\[PubMed\]](#)
68. Amin, A.; Nasim, W.; Mubeen, M.; Sarwar, S.; Urich, P.; Ahmad, A.; Wajid, A.; Khaliq, T.; Rasul, F.; Hammad, H.M. Regional climate assessment of precipitation and temperature in Southern Punjab (Pakistan) using SimCLIM climate model for different temporal scales. *Theor. Appl. Climatol.* **2018**, *131*, 121–131. [\[CrossRef\]](#)
69. Fahad, S.; Li, W.; Lashari, A.H.; Islam, A.; Khattak, L.H.; Rasool, U. Evaluation of land use and land cover Spatio-temporal change during rapid Urban sprawl from Lahore, Pakistan. *Urban Clim.* **2021**, *39*, 100931. [\[CrossRef\]](#)
70. Hussain, S.; Karuppanan, S. Land use/land cover changes and their impact on land surface temperature using remote sensing technique in district Khanewal, Punjab Pakistan. *Geol. Ecol. Landsc.* **2021**, *7*, 46–58. [\[CrossRef\]](#)
71. Imran, Z. *The ‘Press and Pulse’ of Climate Change Strains Farmers in Pakistan*, 10 February 2023 ed.; Bulletin of the Atomic Scientists: Online, 2023.
72. Raziq, A.; Xu, A.; Li, Y.; Zhao, Q. Monitoring of land use/land cover changes and urban sprawl in Peshawar City in Khyber Pakhtunkhwa: An application of geo-information techniques using of multi-temporal satellite data. *J. Remote Sens. GIS* **2016**, *5*, 3. [\[CrossRef\]](#)
73. Garai, S.; Khatun, M.; Singh, R.; Sharma, J.; Pradhan, M.; Ranjan, A.; Rahaman, S.M.; Khan, M.L.; Tiwari, S. Assessing correlation between Rainfall, normalized difference Vegetation Index (NDVI) and land surface temperature (LST) in Eastern India. *Saf. Extrem. Environ.* **2022**, *4*, 119–127. [\[CrossRef\]](#)
74. Gohain, K.J.; Goswami, A.; Mohammad, P.; Kumar, S. Modelling relationship between land use land cover changes, land surface temperature and urban heat island in Indore city of central India. *Theor. Appl. Climatol.* **2023**, *151*, 1981–2000. [\[CrossRef\]](#)
75. Ullah, W.; Ahmad, K.; Ullah, S.; Tahir, A.A.; Javed, M.F.; Nazir, A.; Abbasi, A.M.; Aziz, M.; Mohamed, A. Analysis of the relationship among land surface temperature (LST), land use land cover (LULC), and normalized difference vegetation index (NDVI) with topographic elements in the lower Himalayan region. *Heliyon* **2023**, *9*, e13322. [\[CrossRef\]](#) [\[PubMed\]](#)
76. Kumari, B.; Tayyab, M.; Shahfahad; Salman; Mallick, J.; Khan, M.F.; Rahman, A. Satellite-driven land surface temperature (LST) using Landsat 5, 7 (TM/ETM+ SLC) and Landsat 8 (OLI/TIRS) data and its association with built-up and green cover over urban Delhi, India. *Remote Sens. Earth Syst. Sci.* **2018**, *1*, 63–78. [\[CrossRef\]](#)
77. Abutaleb, K.; Ngie, A.; Darwish, A.; Ahmed, M.; Arafat, S.; Ahmed, F. Assessment of urban heat island using remotely sensed imagery over Greater Cairo, Egypt. *Adv. Remote Sens.* **2015**, *4*, 35. [\[CrossRef\]](#)

78. Hamoodi, M.N. *Study Urban Heat Island and Its Effects on Climate of Al-Kut City Using Geomatics Techniques*; University of Technology: Baghdad, Iraq, 2023.
79. Mathew, A.; Sarwesh, P.; Khandelwal, S.; Raja Shekar, P.; Omeiza Alao, J.; Abdo, H.G.; Almohamad, H.; Abdullah Al Dughairi, A. Thermal dynamics of Jaipur: Analyzing urban heat island effects using in-situ and remotely sensed data. *Cogent Eng.* **2023**, *10*, 2269654. [[CrossRef](#)]
80. Guha, S.; Govil, H.; Dey, A.; Gill, N. Analytical study of land surface temperature with NDVI and NDBI using Landsat 8 OLI and TIRS data in Florence and Naples city, Italy. *Eur. J. Remote Sens.* **2018**, *51*, 667–678. [[CrossRef](#)]
81. Malik, M.S.; Shukla, J.P.; Mishra, S. Relationship of LST, NDBI and NDVI using landsat-8 data in Kandaihimmat watershed, Hoshangabad, India. *Indian J. Geo-Mar. Sci.* **2019**, *48*, 25–31.
82. Patz, J.A.; Campbell-Lendrum, D.; Holloway, T.; Foley, J.A. Impact of regional climate change on human health. *Nature* **2005**, *438*, 310–317. [[CrossRef](#)] [[PubMed](#)]
83. Yasir, M.; Hui, S.; Rahman, S.U.; Ilyas, M.; Zafar, A.; Mehmood, A. Estimation of land surface temperature using LANDSAT-8 data-A case study of district Malakand, Khyber Pakhtunkhwa, Pakistan. *J. Lib. Arts Humanit.* **2020**, *1*, 140–148.
84. Liu, J.; Liu, S.; Tang, X.; Ding, Z.; Ma, M.; Yu, P. The response of land surface temperature changes to the vegetation dynamics in the Yangtze River Basin. *Remote Sens.* **2022**, *14*, 5093. [[CrossRef](#)]
85. Tong, S.; Zhang, J.; Bao, Y.; Lai, Q.; Lian, X.; Li, N.; Bao, Y. Analyzing vegetation dynamic trend on the Mongolian Plateau based on the Hurst exponent and influencing factors from 1982–2013. *J. Geogr. Sci.* **2018**, *28*, 595–610. [[CrossRef](#)]
86. Tran, T.V.; Tran, D.X.; Nguyen, H.; Latorre-Carmona, P.; Myint, S.W. Characterising spatiotemporal vegetation variations using LANDSAT time-series and Hurst exponent index in the Mekong River Delta. *Land Degrad. Dev.* **2021**, *32*, 3507–3523. [[CrossRef](#)]

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.